Assignment - 3

(2025140160…)

01/04/26

DYP

Q.1) Find the fourier series expansion of a function $f(x) = x$, in the interval $0 \le x \le 2\pi$ and $f(x+2\pi) = f(x)$

Given $f(x) = x$, $x \in (0, 2\pi)$ — ①

The Fourier series expansion on interval $(0, 2\pi)$ is given by,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx) \quad \text{— ②}$$

Now,

$$a_0 = \frac{1}{\pi}\int_0^{2\pi} f(x).dx$$

$$= \frac{1}{\pi}\int_0^{2\pi} x.dx$$

$$= \frac{1}{\pi}\left[\frac{x^2}{2}\right]_0^{2\pi}$$

$$= \frac{1}{\pi}\left[\frac{(2\pi)^2}{2} - \frac{(0)^2}{2}\right]$$

$$= \frac{1}{\pi}\left[\frac{4\pi^2}{2}\right]$$

$$\boxed{a_0 = 2\pi} \quad \text{— ③}$$

Let's find value of a_n

$$a_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\cos(nx)\,dx$$

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$$= \frac{1}{\pi}\int_0^{2\pi} x\cos(nx)\,dx$$

$$= \frac{1}{\pi}\left\{ x\frac{\sin(nx)}{n} - \left(-\frac{\cos(nx)}{n^2}\right)\right\}_0^{2\pi}$$

$$= \frac{1}{\pi}\left\{ \frac{x\sin(nx)}{n} + \frac{\cos(nx)}{n^2}\right\}_0^{2\pi}$$

$$= \frac{1}{\pi}\left\{\left[\frac{2\pi\sin 2n\pi}{n} + \frac{\cos(2n\pi)}{n^2}\right] - \left[0 + \frac{\cos(0)}{n^2}\right]\right\}$$

$$= \frac{1}{\pi}\left\{\left[0+\frac{1}{n^2}\right] - \left[0+\frac{1}{n^2}\right]\right\}$$

$$= \frac{1}{\pi}\left\{\frac{1}{n^2} - \frac{1}{n^2}\right\}$$

$$= \frac{1}{\pi}\{0\}$$

$\boxed{a_n = 0}$ ——— ③

Now, $b_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\sin(nx)\,dx$

$$= \frac{1}{\pi}\int_0^{2\pi} x\sin(nx)\,dx$$

$$= \frac{1}{\pi}\left\{ x\left(-\frac{\cos(nx)}{n}\right) - \left(-\frac{\sin(nx)}{n^2}\right)\right\}_0^{2\pi}$$

DYP

$$= \frac{1}{\pi} \int_0^{2\pi} x \cos(nx)\, dx$$

$$= \frac{1}{\pi} \left\{ \frac{x \sin(nx)}{n} - \left(- \frac{\cos(nx)}{n^2} \right) \right\}_0^{2\pi}$$

$$= \frac{1}{\pi} \left\{ \frac{x \sin(nx)}{n} + \frac{\cos(nx)}{n^2} \right\}_0^{2\pi}$$

$$= \frac{1}{\pi} \left\{ \left[\frac{2\pi \sin 2n\pi}{n} + \frac{\cos(2n\pi)}{n^2} \right] - \left[0 + \frac{\cos(0)}{n^2} \right] \right\}$$

$$= \frac{1}{\pi} \left\{ \left[0 + \frac{1}{n^2} \right] - \left[0 + \frac{1}{n^2} \right] \right\}$$

$$= \frac{1}{\pi} \left\{ \frac{1}{n^2} - \frac{1}{n^2} \right\}$$

$$= \frac{1}{\pi} \{ 0 \}$$

$\boxed{a_n = 0}$ —— ③

Now, $b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin(nx)\, dx$

$$= \frac{1}{\pi} \int_0^{2\pi} x \sin(nx)\, dx$$

$$= \frac{1}{\pi} \left\{ x \left(- \frac{\cos(nx)}{n} \right) - \left(\frac{-\sin(nx)}{n^2} \right) \right\}_0^{2\pi}$$

DYP

$$= \frac{1}{\pi}\left\{-\frac{x\cos(nx)}{n} + \frac{\sin(nx)}{n^2}\right\}_0^{2\pi}$$

$$= \frac{1}{\pi}\left\{\left[\frac{-2\pi\cos(2n\pi)}{n} + \frac{\sin(2n\pi)}{n^2}\right] - \left[\frac{-0\cos(0)}{n} + \frac{\sin(0)}{n^2}\right]\right\}$$

$$= \frac{1}{\pi}\left\{\left[\frac{-2\pi\cos(2n\pi)}{n} + \frac{\sin(2n\pi)}{n^2}\right]\right\}$$

$$= \frac{1}{\pi}\left\{\frac{-2\pi}{n} + 0\right\}$$

$$= \frac{1}{\cancel{\pi}}\left[\frac{-2\cancel{\pi}}{n}\right]$$

$$= \frac{-2}{n}$$

$$\boxed{\therefore \quad b_n = \frac{-2}{n}} \quad \text{—— } (5)$$

put eq^n (3), (4) + (5) in eq^n (3)

$$\therefore f(x) = \frac{2\pi}{2} + \sum_{n=1}^{\infty}(0)\cos(nx) + \sum_{n=1}^{\infty}\frac{-2}{n}\sin(nx)$$

$$\boxed{\therefore f(x) = \pi - \sum_{n=1}^{\infty}\frac{2}{n}\sin(nx)}$$

(Q2) Find the fourier series expansion of the function $f(x) = x^2$, in the interval $0 \le x \le 2\pi$ and $f(x+2\pi) = f(x)$

Given $f(x) = x^2 \quad x \in (0, 2\pi)$ — ①

The fourier series on $(0, 2\pi)$ is given by,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx) \quad — ②$$

Now, $a_0 = \frac{1}{\pi}\int_0^{2\pi} f(x)\,dx$

$$= \frac{1}{\pi}\int_0^{2\pi} x^2\,dx$$

$$= \frac{1}{\pi}\int_0^{2\pi} x^2\,dx$$

$$= \frac{1}{\pi}\left\{\frac{x^3}{3}\right\}_0^{2\pi}$$

$$= \frac{1}{\pi}\left\{\frac{(2\pi)^3}{3} - \frac{(0)^3}{3}\right\}$$

$$= \frac{8\pi^{\cancel{3}}}{\cancel{\pi}\,3}$$

$$\boxed{a_0 = \frac{8\pi^2}{3}} \quad — ③$$

Then, $a_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\cos(n\pi)\,dx$

DYP

(Q2) Find the Fourier series expansion of the function $f(x) = x^2$, in the interval $0 \le x \le 2\pi$ and $f(x+2\pi) = f(x)$

Given $f(x) = x^2 \qquad x \in (0, 2\pi)$ — ①

The fourier series on $(0, 2\pi)$ is given by,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx) \quad — ②$$

Now, $a_0 = \frac{1}{\pi}\int_0^{2\pi} f(x)dx$

$$= \frac{1}{\pi}\int_0^{2\pi} x^2 dx$$

$$= \frac{1}{\pi}\int_0^{2\pi} x^2 dx$$

$$= \frac{1}{\pi}\left\{\frac{x^3}{3}\right\}_0^{2\pi}$$

$$= \frac{1}{\pi}\left\{\frac{(2\pi)^3}{3} - \frac{(0)^3}{3}\right\}$$

$$= \frac{8\pi^{\cancel{3}}}{\cancel{\pi}3}$$

$$\boxed{a_0 = \frac{8\pi^2}{3}} \quad — ③$$

Then, $a_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\cos(n\pi)dx$

DYP

$$= \frac{1}{\pi} \int_0^{2\pi} x^2 \cos(nx)\, dx$$

$$= \frac{1}{\pi} \left\{ x^2 \left(\frac{\sin(nx)}{n} \right) - 2x \left(-\frac{\cos(nx)}{n^2} \right) + 2 \left(\frac{-\sin(nx)}{n^3} \right) \right\}_0^{2\pi}$$

$$= \frac{1}{\pi} \left\{ \frac{x^2 \sin(nx)}{n} + \frac{2x \cos(nx)}{n^2} - \frac{2\sin(nx)}{n^3} \right\}_0^{2\pi}$$

$$= \frac{1}{\pi} \left\{ \left[\frac{2\pi^2 \sin(2n\pi)}{n} + \frac{4\pi \cos(2n\pi)}{n^2} - \frac{2\sin(2n\pi)}{n^3} \right] - \left[\frac{0^2 \sin(0)}{n} + \frac{2(0)\cos(0)}{n^2} - \frac{2\sin(0)}{n^3} \right] \right\}$$

$$= \frac{1}{\pi} \left\{ \left[0 + \frac{4\pi}{n^2} - 0 \right] - \left[0 + 0 - 0 \right] \right\}$$

$$= \frac{1}{\cancel{\pi}} \left[\frac{4\cancel{\pi}}{n^2} \right]$$

$$\boxed{a_n = \frac{4}{n^2}} \quad \text{—— } ④$$

Now, $b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin(nx)\, dx$

$$= \frac{1}{\pi} \int_0^{2\pi} x^2 \sin(nx)\, dx$$

$$= \frac{1}{\pi} \left\{ x^2 \left(-\frac{\cos(nx)}{n} \right) - 2x \left(-\frac{\sin(nx)}{n^2} \right) + 2 \frac{\cos(nx)}{n^3} \right\}_0^{2\pi}$$

$$= \frac{1}{\pi}\left\{ \frac{-x^2\cos(nx)}{n} + \frac{2x\sin(nx)}{n^2} + \frac{2\cos(nx)}{n^3} \right\}_0^{2\pi}$$

$$= \frac{1}{\pi}\left\{ \left[\frac{-4\pi^2}{n} + 0 + \frac{2}{n^3}\right] - \left[0 + 0 + \left[\frac{2}{n^3}\right]\right] \right\}$$

$$= \frac{1}{\pi}\left\{ \frac{-4\pi^2}{n} + \cancel{\frac{2}{n^3}} - \cancel{\frac{2}{n^3}} \right\}$$

$$= \frac{1}{\cancel{\pi}}\left[-\frac{4\pi^2}{n} \right]$$

$$= \frac{-4\pi}{n}$$

$$\therefore \boxed{b_n = \frac{-4\pi}{n}} \quad \text{——— (5)}$$

Put (3), (4) & (5) in eq^n (2)

$$f(x) = \frac{8\pi^2}{3} \times \frac{1}{2} + \sum_{n=1}^{\infty} \frac{4}{n^2}\cos(nx) + \sum_{n=1}^{\infty}\left(\frac{-4\pi}{n}\right)\sin(nx)$$

$$\therefore \boxed{f(x) = \frac{4\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2}\cos(nx) - \sum_{n=1}^{\infty} \frac{4\pi}{n}\sin(nx)}$$

(Q3) Find the fourier series expansion of function $f(x) = |x|$, in the interval $-\pi \le x \le \pi$ & $f(x+2\pi) = f(x)$

Now, interval is $(-\pi, \pi)$

let's check the function.

$f(x) = |x|$ ——— ①

Put $x = (-x)$

$f(-x) = |-x|$

$f(-x) = f(x)$

$\therefore$ $f(x) = |x|$ is even function.

Now, Fourier series expansion for interval $(-\pi, \pi)$ is.

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx)$$

Here, for even function,

$$a_0 = \frac{2}{\pi}\int_0^{\pi} f(x)\,dx \;,\; a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos(nx)\,dx \;,\; b_n = 0$$

$$\therefore \; f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) \quad \text{——— ②}$$

Now, $a_0 = \frac{2}{\pi}\int_0^{\pi} f(x)\cdot dx$

$$a_0 = \frac{2}{\pi}\int_0^{\pi} x \cdot dx$$

$$a_0 = \frac{2}{\pi}\left[\frac{x^2}{2}\right]_0^{\pi}$$

$$= \frac{2}{\pi}\left[\frac{\pi^2}{2} - 0\right]$$

$a_0 = \pi$ —— (3)

Now, $a_n = \frac{2}{\pi}\int_0^{\pi} x\cos(nx)$

$$= \frac{2}{\pi}\left\{ x\frac{\sin(nx)}{n} - 1\left(-\frac{\cos(nx)}{n^2}\right)\right\}_0^{\pi}$$

$$= \frac{2}{\pi}\left\{ \frac{x\sin(nx)}{n} + \frac{\cos(nx)}{n^2}\right\}_0^{\pi}$$

$$= \frac{2}{\pi}\left\{\left[\frac{\pi\sin(n\pi)}{n} + \frac{\cos(n\pi)}{n^2}\right] - \left[\frac{0\sin(n0)}{n} + \frac{\cos(0n)}{n^2}\right]\right\}$$

$$= \frac{2}{\pi}\left\{\left[0 + \frac{(-1)^n}{2}\right] - \left[\frac{1}{n^2}\right]\right\}$$

$$= \frac{2}{\pi}\left[\frac{(-1)^n}{n^2} - \frac{1}{n^2}\right]$$

$$\boxed{a_n = \frac{2[(-1)^n - 1]}{\pi n^2}} \quad \text{—— (4)}$$

Put (3) + (4) in (2)

$$\therefore \boxed{f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2[(-1)^n - 1]}{\pi n^2}\cos(nx)}$$

DYP ④

(Q4) Find the fourier series expansion of function $f(x) = x^3$ in interval $(-\pi \le x \le \pi)$ and $f(x+2\pi) = f(x)$

Given $f(x) = x^3 \quad \forall \; x \in (-\pi, \pi)$ —— ①

As interval is $(-\pi, \pi)$ then, check the $f(x)$ is odd / even.

$f(x) = x^3$

Put $x = (-x)$

$f(-x) = -x^3$

$f(-x) = -f(x)$

$\therefore \quad f(x) = x^3$ is Odd function.

So, fourier series of Odd function on $(-\pi, \pi)$ is

$$f(x) = \sum_{n=1}^{\infty} b_n \sin(nx) \quad \text{—— ②}$$

$$\text{So, } b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin(nx)\, dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x^3 \sin(nx)\, dx$$

$$= \frac{2}{\pi} \left\{ x^3 \left(\frac{-\cos(nx)}{n} \right) - 3x^2 \left(-\frac{\sin(nx)}{n^2} \right) + 6x \left(\frac{\cos(nx)}{n^3} \right) - \frac{6\sin(nx)}{n^4} \right\}_0^{\pi}$$

$$= \frac{2}{\pi} \left\{ \frac{-x^3 \cos(nx)}{n} + \frac{3x^2 \sin(nx)}{n^2} + \frac{6x\cos(nx)}{n^3} - \frac{6\sin(nx)}{n^4} \right\}_0^{\pi}$$

DYP

$$= \frac{2}{\pi}\left\{\left[\frac{-\pi^3\cos(n\pi)}{\pi} + \frac{3\pi^2\sin(n\pi)}{n^2} + \frac{6\pi\cos(n\pi)}{n^3} - \frac{6\sin(n\pi)}{n^4}\right]\right.$$

$$\left. - \left[\frac{0\cos(0x)}{n} + \frac{3(0)\sin(n0)}{n^2} + \frac{6(0)\cos(n0)}{n^3} - \frac{6\sin(n0)}{n^4}\right]\right\}$$

$$= \frac{2}{\pi}\left\{\left[-\frac{\pi^3(-1)^n}{n} + 0 + \frac{6\pi(-1)^n}{n^3} - 0\right] - \left[0 + 0 + 0 - 0\right]\right\}$$

$$= \frac{2}{\pi}\left\{\frac{-\pi^3(-1)^n}{n} + \frac{6\pi(-1)^n}{n^3}\right\}$$

$$b_n = \frac{2(-1)^n}{n}\left(\frac{6}{n^2} - \pi^2\right) \longrightarrow (3)$$

Put eq^n (3) in eq^n (2)

$$\therefore f(x) = \sum_{n=1}^{\infty} \frac{2(-1)^n}{n}\left(\frac{6}{n^2} - \pi^2\right) \cancel{\cos} \sin(nx)$$

$$\therefore \boxed{f(x) = \sum_{n=1}^{\infty} \frac{2(-1)^n}{n}\left(\frac{6}{n^2} - \pi^2\right) \sin(nx)}$$

(Q.5) Find the fourier series expansion of function $f(x) = 4 - x^2$ in interval $0 \le x \le 2$ & $f(x+2) = f(x)$

Given interval is $[0, 2]$

$$2l = 2$$
$$l = 1$$

In fourier series on interval $(0, 2)$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n\pi x) + \sum_{n=1}^{\infty} b_n \sin(n\pi x) \quad \text{—— ①}$$

Now, $a_0 = \frac{1}{l}\int_0^{2l} f(x)\,dx$

$$a_0 = 1\int_0^2 (4 - x^2)\,dx$$

$$a_0 = \left[4x - \frac{x^3}{3}\right]_0^2$$

$$a_0 = 8 - 8/3$$

$$\boxed{a_0 = 16/3} \quad \text{—— ②}$$

Now, $a_n = \frac{1}{l}\int_0^{2l} f(x)\cos(n\pi x)\,dx$

$$= 1\int_0^2 (4 - x^2)\cos(nx\pi)\,dx$$

Here, by using Bernoulli's theorem,

DYP

$$a_n = \left\{ (4-x^2)\left(\frac{\sin(n\pi x)}{n\pi}\right) - (-2x)\left(-\frac{\cos(n\pi x)}{n^2\pi^2}\right) + (-2)\left(\frac{-\sin(n\pi x)}{n^3\pi^3}\right) \right\}_0^2$$

$$= \left\{ 4-x^2\left(\frac{\sin(n\pi x)}{n\pi}\right) - \frac{2x\cos(n\pi x)}{n^2\pi^2} + \frac{2\sin(n\pi x)}{n^3\pi^3} \right\}_0^2$$

$$= \left\{ \left[\frac{4-(2)^2\sin(n2\pi)}{n\pi} - \frac{4\cos(2n\pi)}{n^2\pi^2} + \frac{2\sin(2n\pi)}{n^3\pi^3} \right] - \right.$$

$$\left. \left[\frac{4-(0)^2\sin(0n\pi)}{n\pi} - \frac{2(0)\cos(2n\pi)}{n^2\pi^2} + \frac{2\sin(0n\pi)}{n^3\pi^3} \right] \right\}$$

$$= \left\{ \left[0 - \frac{4}{n^2\pi^2} + 0 \right] - \left[0+0+0 \right] \right\}$$

$$\boxed{a_n = \frac{-4}{n^2\pi^2}} \quad \text{---} \ (3)$$

Now, $b_n = \frac{1}{l}\int_0^{2l} f(x)\sin\left(\frac{n\pi x}{l}\right)dx$

$$= 1\int_0^2 (4-x^2)\sin(n\pi x)\,dx$$

$$b_n = \left\{ 4-x^2\left(\frac{-\cos(n\pi x)}{n\pi}\right) - (-2x)\left(-\frac{\sin(n\pi x)}{n^2\pi^2}\right) + (-2)\left(\frac{\cos(n\pi x)}{n^3\pi^3}\right) \right\}_0^2$$

$$b_n = \left\{ \frac{-(4-x^2)\cos(n\pi x)}{n\pi} - \frac{2x\sin(n\pi x)}{n^2\pi^2} - \frac{2\cos(n\pi x)}{n^3\pi^3} \right\}_0^2$$

$$b_n = \left\{ \left[0-0-\frac{2(1)}{n^3\pi^3} \right] - \left[\frac{-4(1)}{n\pi} - 0 - \left(\frac{2(1)}{n^3\pi^3}\right) \right] \right.$$

DYP

$$bn = \cancel{\frac{-2}{n^3\pi^3}} + \frac{4}{n\pi} + \cancel{\frac{2}{n^3\pi^3}}$$

$$\boxed{bn = \frac{4}{n\pi}} \quad \text{——— } ④$$

Put ②, ③ & ④ in ①

$$\boxed{f(x) = \frac{8}{3} - \sum_{n=1}^{\infty} \frac{4}{n^2\pi^2} \cos(n\pi x) + \sum_{n=1}^{\infty} \frac{4}{n\pi} \sin(n\pi x)}$$

(Q.6) Find the fourier series expansion of function $f(x) = 2x - x^2$; $x \in (0,3)$ and Period is 3.

Here, $2L = 3$

$L = 3/2$

$\therefore$ The fourier series on interval $(0,3)$ is given by,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{2n\pi x}{3}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{2n\pi x}{3}\right) \quad - ①$$

$$a_0 = \frac{2}{3} \int_0^3 f(x)\, dx$$

$$= \frac{2}{3} \int_0^3 (2x - x^2)\, dx$$

$$= \frac{2}{3} \left\{ \frac{\cancel{2}x^2}{\cancel{2}} - \frac{x^3}{3} \right\}_0^3$$

$$= \frac{2}{3} \left\{ [9-9] - [0-0] \right\}$$

$$a_0 = 0 \quad \text{——— } ②$$

$$a_n = \frac{2}{3}\int_0^3 f(x)\cos\left(\frac{2n\pi x}{3}\right)dx$$

$$= \frac{2}{3}\int_0^3 (2x-x^2)\cos\left(\frac{2n\pi x}{3}\right)dx$$

$$= \frac{2}{3}\left\{ (2x-x^2)\frac{\left(+\sin\left(\frac{2n\pi x}{3}\right)\right)}{\left(\frac{2n\pi}{3}\right)} - (2-2x)\left(\frac{-\cos\left(\frac{2n\pi x}{3}\right)}{\left(\frac{2n\pi}{3}\right)^2}\right) + (-2)\left(\frac{-\sin\left(\frac{2n\pi x}{3}\right)}{\left(\frac{2n\pi}{3}\right)^3}\right)\right\}_0^3$$

$$= \frac{2}{3}\left\{\left[(6-9)\frac{\left(\sin\left(\frac{6n\pi}{3}\right)\right)}{\left(\frac{2n\pi}{3}\right)} + (2-6)\frac{\left(\cos\left(\frac{6n\pi}{3}\right)\right)}{\left(\frac{2n\pi}{3}\right)^2} + \frac{2\sin\left(\frac{6n\pi x}{3}\right)}{\left(\frac{2n\pi}{3}\right)^3}\right]\right.$$

$$\left. - \left[0 + 2\left(\frac{\cos(0)}{\left(\frac{2n\pi}{3}\right)^2}\right) + \frac{2\sin(0)}{\left(\frac{2n\pi}{3}\right)^3}\right]\right\}$$

$$= \frac{2}{3}\left\{\left[0 + \frac{2-6}{\left(\frac{2n\pi}{3}\right)^2} + 0\right] - \left[0 + \frac{2}{\left(\frac{2(n\pi)}{3}\right)^2} + 0\right]\right\}$$

$$= \frac{2}{3}\left\{\frac{-4}{\left(\frac{2n\pi}{3}\right)^2} - \frac{2}{\left(\frac{2n\pi}{3}\right)^2}\right\}$$

$$= \frac{2}{\cancel{3}}\left\{\frac{-\cancel{6}^{\,2}}{\left(\frac{2n\pi}{3}\right)^2}\right\} \longrightarrow \frac{-4}{\left(\frac{2n\pi}{3}\right)^2}$$

$$-4 \times \left(\frac{3}{2n\pi}\right)^2$$

$$= \frac{-\cancel{4} \times 9}{\cancel{4} n^2 \pi^2}$$

$$a_n = \frac{-9}{n^2\pi^2} \quad \text{—— ③}$$

Now, $b_n = \frac{2}{3}\int_0^3 f(x)\sin\left(\frac{2n\pi x}{3}\right)dx$

$$b_n = \frac{2}{3}\int_0^3 (2x - x^2)\sin\left(\frac{2n\pi x}{3}\right)dx$$

$$= \frac{2}{3}\left\{(2x-x^2)\left(\frac{-\cos\left(\frac{2n\pi x}{3}\right)}{\frac{2n\pi}{3}}\right) - (2-2x)\left(\frac{-\sin\left(\frac{2n\pi x}{3}\right)}{\left(\frac{2n\pi}{3}\right)^2}\right) + \left((-2)\left(\frac{\cos\left(\frac{2n\pi x}{3}\right)}{\left(\frac{2n\pi}{3}\right)^3}\right)\right)\right\}_0^3$$

After solving this we get b_n,

$$b_n = -\frac{3}{\pi n} \quad \text{—— ④}$$

Put ②, ③ + ④ in eq^n ①

$$\boxed{f(x) = -\sum_{n=1}^{\infty}\frac{9}{n^2\pi^2}\cos\left(\frac{2n\pi x}{3}\right) - \sum_{n=1}^{\infty}\frac{3}{n\pi}\sin\left(\frac{2n\pi x}{3}\right)}$$

(Q7) Find the fourier series expansion of the function $f(x) = x^2$, in the interval $(-1 \le x \le 1)$ and $f(x+2) = f(x)$

$f(x) = x^2$

Put $x = (-x)$

$f(-x) = x^2$

$\therefore$ $f(x) = x^2$ is even function

The fourier series of even function on $(-1,1)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n\pi x) \quad \text{—— ①}$$

Now, $a_0 = \frac{2}{l}\int_0^l f(x)\,dx$

$$a_0 = 2\int_0^1 x^2\,dx$$

$$a_0 = 2\left[\frac{x^3}{3}\right]_0^1$$

$$a_0 = 2\left(\frac{1}{3} - 0\right)$$

$$\boxed{a_0 = 2/3} \quad \text{—— ②}$$

$$a_n = \frac{2}{l}\int_0^l f(x)\cos(n\pi x)\,dx$$

$$= 2\int_0^1 x^2 \cos(n\pi x)\,dx$$

By using Bernoulli's Rules

DYP

(Q7) Find the fourier series expansion of the function $f(x) = x^2$, in the interval $(-1 \le x \le 1)$ and $f(x+2) = f(x)$

$f(x) = x^2$

Put $x = (-x)$

$f(-x) = x^2$

$\therefore f(x) = x^2$ is even function

The fourier series of even function on $(-1,1)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n\pi x) \quad \text{--- (1)}$$

Now, $a_0 = \frac{2}{l}\int_0^l f(x)\,dx$

$$a_0 = 2\int_0^1 x^2\,dx$$

$$a_0 = 2\left[\frac{x^3}{3}\right]_0^1$$

$$a_0 = 2\left(\frac{1}{3} - 0\right)$$

$$\boxed{a_0 = 2/3} \quad \text{--- (2)}$$

$$a_n = \frac{2}{l}\int_0^l f(x)\cos(n\pi x)\,dx$$

$$= 2\int_0^1 x^2\cos(n\pi x)\,dx$$

By using Bernoulli's Rules

$$= 2\left\{\left[x^2\left(\frac{\sin(n\pi x)}{n\pi}\right) - 2x\left(\frac{-\cos(n\pi x)}{n^2\pi^2}\right) + 2\left(\frac{-\sin(n\pi x)}{n^3\pi^3}\right)\right]\right\}_0^1$$

$$= 2\left[\left(0 + \frac{2(-1)^n}{n^2\pi^2} + 0\right) - (0 + 0 - 0)\right]$$

$$a_n = \frac{4(-1)^n}{n^2\pi^2} \quad \text{——} \quad ④$$

Put ④ & ③ in eqⁿ ①

∴ ~~f(x)~~

$$\boxed{f(x) = \frac{1}{3} + \sum_{n=1}^{\infty} \frac{4(-1)^n}{n^2\pi^2}\cos(n\pi x)}$$

(Q.8) Find the half range cosine series for the function $f(x) = (\pi - x)$, $0 < x < \pi$

$f(x) = (\pi - x) \quad \forall \quad 0 \le x \le \pi$ —— ①

Cosine Series for $(0, \pi)$ is given by,

$$a_0 = \frac{2}{\pi}\int_0^{\pi} (\pi - x)\, dx$$

$$= \frac{2}{\pi}\left[\pi x - \frac{x^2}{2}\right]_0^{\pi}$$

$$a_0 = \frac{2}{\pi}\left\{\left[\pi^2 - \frac{\pi^2}{2}\right] - [0 - 0]\right\}$$

$$a_0 = \frac{\cancel{2}}{\cancel{\pi}}\left(\frac{\pi^{\cancel{2}}}{\cancel{2}}\right) = \pi$$

DYP

$a_0 = \pi \quad \text{---} \quad (2)$

Now, $a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos(nx)\, dx$

$$= \frac{2}{\pi} \int_0^{\pi} (\pi - x) \cos(nx)\, dx$$

$$= \frac{2}{\pi} \left\{ (\pi - x)\left(\frac{\sin(nx)}{n} \right) - (-1)\left(\frac{-\cos(nx)}{n^2} \right) \right\}_0^{\pi}$$

$$= \frac{2}{\pi} \left\{ (\pi - x) \frac{\sin(nx)}{n} - \frac{\cos(nx)}{n^2} \right\}_0^{\pi}$$

$$= \frac{2}{\pi} \left\{ \left[0 \frac{\sin(n\pi)}{n} - \frac{\cos(n\pi)}{n^2} \right] - \left[\pi \cdot 0 \frac{\sin(0n)}{n} - \frac{\cos(0n)}{n^2} \right] \right\}$$

$$= \frac{2}{\pi} \left\{ \left[0 - \frac{(-1)^n}{n^2} \right] - \left[0 - \frac{1}{n^2} \right] \right\}$$

$$= \frac{2}{\pi} \left\{ \frac{-(-1)^n}{n^2} + \frac{1}{n^2} \right\}$$

$$a_n = \frac{2}{n^2 \pi} \left\{ 1 - (-1)^n \right\}$$

$$\therefore \boxed{f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{n^2 \pi} (1 - (-1)^n) \cos(nx)}$$

DYP

(Q.9) Find the half range sine series for function $f(x) = \pi^2 - x^2, \ 0 < x < \pi$

Formula for Half Range Sine series.

$$f(x) = \sum_{n=1}^{\infty} b_n \sin(nx)$$

where. $b_n = \frac{2}{\pi}\int_0^{\pi} f(x) \sin(nx)\,dx$

$$= \frac{2}{\pi}\int_0^{\pi} (\pi^2 - x^2) \sin(nx)\,dx$$

$$= \frac{2}{\pi}\left\{(\pi^2-x^2)\left(-\frac{\cos(nx)}{n}\right) - (-2x)\left(-\frac{\sin(nx)}{n^2}\right) + (-2)\left(\frac{\cos(nx)}{n^3}\right)\right\}_0^{\pi}$$

$$= \frac{2}{\pi}\left\{\left[-(\pi^2-x^2)\left(\frac{\cos(nx)}{n}\right) - \frac{2x\sin(nx)}{n^2} - \frac{2\cos(nx)}{n^3}\right]_0^{\pi}\right\}$$

$$= \frac{2}{\pi}\left\{\left[0 - \frac{2\pi\sin(n\pi)}{n^2} - \frac{2\cos(n\pi)}{n^3}\right] - \left[\frac{-\pi^2\cos(0n)}{n} - 0 - \frac{2\cos(n0)}{n^3}\right]\right\}$$

$$= \frac{2}{\pi}\left\{\left[0 - 0 - \frac{2(-1)^n}{n^3}\right] - \left[\frac{-\pi^2(1)}{n} - \frac{2(1)}{n^3}\right]\right\}$$

$$= \frac{2}{\pi}\left[\frac{-2(-1)^n}{n^3} + \frac{\pi^2}{n} + \frac{2}{n^3}\right]$$

$\therefore$ Answer will be,

$$\boxed{f(x) = \sum_{n=1}^{\infty} \frac{2}{\pi}\left(\frac{-2(-1)^n}{n^3} + \frac{\pi^2}{n} + \frac{2}{n^3}\right)\sin(nx)}$$

(Q.10) Find the half range cosine series for the function.
$f(x) = x - x^2 , \; 0 \leq x \leq 1$

Formula for half range Cosine series is.

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) \quad \text{--- (1)}$$

Here, $a_0 = \frac{2}{l} \int_0^l f(x)\,dx$

$$= 2 \int_0^1 (x - x^2)\,dx$$

$$= 2 \left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1$$

$$= 2 \left[\frac{1}{2} - \frac{1}{3}\right]$$

$$= 2 \left[\frac{3-2}{6}\right]$$

$$= 2 \left[\frac{1}{6}\right]$$

$$\boxed{a_0 = \frac{1}{3}} \quad \text{--- (2)}$$

$a_n = \frac{2}{l} \int_0^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx \qquad [\because l = 1]$

$$= 2 \int_0^1 (x - x^2) \cos(n\pi x)\,dx$$

DYP

$$= 2\left\{\left[(x-x^2)\left(\frac{\sin(n\pi x)}{n\pi}\right) - (1-2x)\left(\frac{-\cos(n\pi x)}{n^2\pi^2}\right) + (-2)\left(\frac{-\sin(n\pi x)}{n^3\pi^3}\right)\right]_0^1\right\}$$

$$= 2\left[(x-x^2)\frac{\sin(n\pi x)}{n\pi} + (1-2x)\frac{\cos(n\pi x)}{n^2\pi^2} + \frac{2\sin(n\pi x)}{n^3\pi^3}\right]_0^1$$

$$= 2\left\{\left[0 + \frac{(-1)(-1)^n}{n^2\pi^2} + 0\right] - \left[0 + \frac{(-1)^n}{n^2\pi^2} + 0\right]\right\}$$

$$= 2\left\{\frac{-(-1)^n}{n^2\pi^2} - \frac{1}{n^2\pi^2}\right\}$$

$$= \frac{-2}{n^2\pi^2}\left((-1)^n + 1\right)$$

$$\therefore \boxed{f(x) = \frac{1}{6} - \sum_{n=1}^{\infty} \frac{2}{n^2\pi^2}\left(1+(-1)^n\right)\cos(n\pi x)}$$

(Q.11) Find the Half Range sine series for the function

$f(x) = e^x$, $0 < x < 1$

$f(x) = e^x$ ——— ①

Half Range Sine Series is given by.

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$$

here: $(l = 1)$

$$b_n = \frac{2}{l}\int_0^l f(x)\sin\left(\frac{n\pi x}{l}\right)dx$$

DYP

$$b_n = 2\int_0^1 e^x \sin(n\pi x)\,dx$$

$$= 2\left[\frac{e^x}{1^2+n^2\pi^2}\left(\sin(n\pi x) - b\cos(n\pi x)\right)\right]_0^1$$

$$= 2\left\{\left[\frac{e^1}{1^2+(n\pi)^2}\left(\sin(n\pi) - (n\pi)\cos(n\pi)\right)\right] - \left[\frac{e^0}{1^2+(n\pi)^2}\sin(0n\pi) - \pi n\cos(0)\right]\right\}$$

$$= 2\left\{\left[\frac{e}{1^2+(n\pi)^2}\times\left(0 - n\pi(-1)^n\right)\right] - \left[\frac{1}{1^2+(n\pi)^2}\left(0 - \pi n(1)\right)\right]\right\}$$

$$= \frac{2n\pi}{1+n^2\pi^2}\left[1 - e(-1)^n\right]$$

$$\therefore \boxed{f(x) = \sum_{n=1}^{\infty}\frac{2n\pi}{1+n^2\pi^2}\left(1 - e(-1)^n\right)\sin(n\pi x)}$$

(Q.12) Obtain the constant term, coefficient of 1st Cosine & Sine terms in the Fourier series for y, where y is given in following table.

x	0	1	2	3	4	5	6
y	9	18	24	28	26	20	9

from here series will repeat

$\therefore 2l = 6$

$(l = 3)$

$\therefore$ The formula for Harmonic Analysis is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n\cos\left(\frac{n\pi x}{l}\right) + b_n\sin\left(\frac{n\pi x}{l}\right)\right]$$

DYP

$\therefore f(x) = \frac{a_0}{2} + a_1 \cos\left(\frac{x\pi}{3}\right) + b_1 \sin\left(\frac{\pi x}{3}\right) + \text{---} \quad \text{—} \ ①$

$a_0 = 2 \times (\text{mean of } y \text{ on } (0,6))$

$a_1 = 2 \times (\text{mean of } y\cos\left(\frac{\pi x}{3}\right) \text{ on } (0,6))$

$b_1 = 2 \times (\text{mean of } y\sin\left(\frac{\pi x}{3}\right) \text{ on } (0,6))$

x	y	$\cos\left(\frac{x\pi}{3}\right)$	$y\cos\left(\frac{x\pi}{3}\right)$	$\sin\left(\frac{x\pi}{3}\right)$	$y\sin\left(\frac{x\pi}{3}\right)$
0	9	1	9	0	0
1	18	0.5	9	0.866	15.588
2	24	-0.5	-12	0.866	20.788
3	28	-1	-28	0	0
4	26	-0.5	-13	-0.866	-22.516
5	20	0.5	10	-0.866	-17.32

Now, $a_0 = 2 \times \frac{\Sigma y}{6} \rightarrow 2 \times \frac{125}{6} = 41.66$

$a_1 = 2 \times \frac{\Sigma y\cos\left(\frac{\pi x}{3}\right)}{6} = -8.33$

$b_1 = 2 \times \frac{\Sigma y\sin\left(\frac{\pi x}{3}\right)}{6} = \frac{2 \times (-3.46)}{6} = 1.153$

Put values of a_0, a_1, b_1 in eqⁿ ①

$\therefore f(x) = \frac{(41.66)}{2} + (-8.33)\left(\cos\left(\frac{\pi x}{3}\right)\right) + \left(1.153 \sin\left(\frac{\pi x}{3}\right)\right) \text{---}$

$$\boxed{\therefore f(x) = 20.83 - 8.33\cos\left(\frac{\pi x}{3}\right) + 1.153\sin\left(\frac{\pi x}{3}\right)}$$

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